

Isobaric Vapor–Liquid Equilibria for Ternary Systems Acetone + Ethanol + 1-Butanol and Ethanol + 1-Butanol + 2-Ethyl-1-hexanol and Their Constituent Binary Systems

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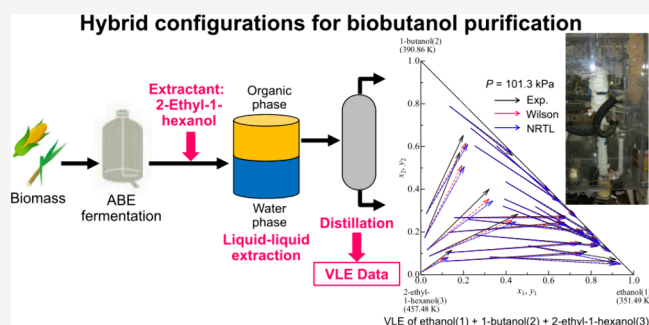
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ABSTRACT: Experimental isobaric vapor–liquid equilibrium (VLE) data at 101.3 kPa were obtained for the development of purification processes for biobutanol. We focused on hybrid configurations using a combination of liquid–liquid extraction and distillation alternatives for the biobutanol purification, i.e., the separation of the product into raffinate and extract phases using liquid–liquid extraction and distillation for the recovery of the extractant, acetone, butanol, and ethanol. We used 2-ethyl-1-hexanol as the extractant for liquid–liquid extraction. Two ternary systems, i.e., acetone + ethanol + 1-butanol and ethanol + 1-butanol + 2-ethyl-1-hexanol, and their six constituent binary systems were studied at 101.3 kPa. The binary VLE data were represented by using the Wilson, NRTL, and UNIQUAC models. The ternary VLE data were compared with the predicted data using the binary parameters of these models obtained from the experimental VLE data of the constituent binary systems, and good prediction accuracy was obtained. Two group contribution methods, i.e., modified UNIFAC (Dortmund) and ASOG, were also used for VLE prediction for the binary and ternary systems.



1. INTRODUCTION

Recently, the use of petroleum refineries that use fossil fuels, mainly crude oil, has not only led to the depletion of fossil resources as a carbon source but also resulted in serious environmental problems, such as global warming and climate change caused by CO₂ emissions.^{1,2} To address these issues, biorefining technologies to use biofuels have been developed to realize a sustainable carbon source, resulting not only in reducing CO₂ emissions but also in reducing the risk for oil price fluctuations.³ Biobutanol has attracted attention as a biofuel because, compared with bioethanol, biobutanol has a higher energy density, lower vapor pressure, and higher flash point and is less corrosive, less hygroscopic, and less miscible with water.^{3–10} In addition, biobutanol has better miscibility with gasoline, compared with bioethanol, which means that biobutanol can be blended in any concentration with gasoline or used as a pure fuel without any modification to the engine being required. In particular, biobutanol has 30% more energy content (27.83 MJ/L) than bioethanol (21.27 MJ/L).¹¹

Biobutanol is obtained from renewable biomass resources, such as rice straw and corn fiber, by fermentation with *Clostridium* bacteria. The process is known as acetone–butanol–ethanol (ABE) fermentation because the end products are primarily acetone, 1-butanol, and ethanol, with a mass ratio of 3:6:1.^{6,7,9,10,12–15} However, the 1-butanol concentration in the

product obtained by ABE fermentation is only (0.5–2) wt %, ^{1,8,16,17} because 1-butanol is toxic to microorganisms, which causes the microorganisms to die off as 1-butanol accumulates in the fermentation broth.^{7,18} In addition, the presence of homogeneous ethanol–water and heterogeneous 1-butanol–water azeotropes makes the separation difficult.^{19,20} Therefore, the development of an energy-efficient process for the purification of biobutanol is highly desirable. Several purification methods for biobutanol, including distillation, adsorption, liquid–liquid extraction, pervaporation, gas stripping, perstraction, and reverse osmosis, have been proposed. Details of these separation techniques have been summarized in several reviews^{1,6,7,13,14,21} and books.^{9,10} More recently, freeze crystallization for 1-butanol separation has been also proposed.^{22,23} Among these methods, membrane-based systems show a high selectivity in the purification and have lower energy requirements. However, membrane-based sys-

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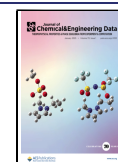


Table 1. Chemicals Used in this Work

Chemical	CAS Registry Number	Source	Grade	Purification method	Mass-fraction purity	Analytical method	Water content (mass ppm)
Acetone	67-64-1	FUJIFILM Wako Pure Chemical Co.	Special grade	Molecular sieves 3A	0.999	GC	—
Ethanol	64-17-5	FUJIFILM Wako Pure Chemical Co.	Special grade	Molecular sieves 3A	0.999	GC	39
1-Butanol	71-36-3	FUJIFILM Wako Pure Chemical Co.	Special grade	Molecular sieves 3A	0.999	GC	17
2-Ethyl-1-hexanol	104-76-7	FUJIFILM Wako Pure Chemical Co.	Special grade	Molecular sieves 4A	0.999	GC	35

Table 2. Normal Boiling Points (T_b), Densities (ρ) at 298.15 K, and Liquid Molar Volumes (v_l^L) at 298.15 K of Pure Components Used in This Work

Component	T_b (K) ^a		ρ (298.15 K) (kg m ⁻³) ^c		$v_l^L \times 10^{-6}$ (m ³ mol ⁻¹)
	Experimental	Literature ^b	Experimental	Literature ^b	
Acetone	329.23	329.22	784.75	784.40	74.01
Ethanol	351.45	351.45	785.10	784.93	58.68
1-Butanol	390.89	390.88	805.74	805.75	91.99
2-Ethyl-1-hexanol	457.46	457.49	832.41	829.0	156.45

^aAt $P = 101.3$ kPa, its corrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. Standard uncertainties are $u(T_b) = 0.06$ K. ^bReference 72. ^cUnder atmospheric pressure. Standard uncertainties are $u(\rho) = 0.11$ kg·m⁻³, $u(T) = 0.05$ K.

Table 3. Determined Antoine Equation Constants^a of Pure Components Used in This Work

Component	A_i	B_i	C_i	$ P_{i,\text{exptl}}^s - P_{i,\text{calcd}}^s / P_{i,\text{exptl}}^s \times 100$	Literature (Antoine constant)	$ P_{i,\text{lit}}^s - P_{i,\text{calcd}}^s / P_{i,\text{lit}}^s \times 100$	Literature (Vapor pressure)
Acetone	6.28028	1231.603	-41.091	0.21	88	0.28	74
Ethanol	7.29043	1630.479	-42.848	0.42	88	0.95	78
1-Butanol	6.63293	1389.635	-90.256	0.36	88	0.50	78
2-Ethyl-1-hexanol	6.06296	1358.827	-122.578	0.37	72	0.61	49

^a $\log(P_i^s/\text{kPa}) = A_i - B_i/[(T/\text{K}) + C_i]$. ^b $|P_{i,\text{exptl}}^s - P_{i,\text{calcd}}^s / P_{i,\text{exptl}}^s| \times 100 = (100/\text{NDP}) \sum_{k=1}^{\text{NDP}} (P_{i,\text{exptl}}^s - P_{i,\text{calcd}}^s / P_{i,\text{exptl}}^s)$, where $P_{i,\text{exptl}}^s$ is the experimental vapor pressure and $P_{i,\text{calcd}}^s$ is the calculated vapor pressure using Antoine constants in the literature. ^c $|P_{i,\text{lit}}^s - P_{i,\text{calcd}}^s / P_{i,\text{lit}}^s| \times 100 = (100/\text{NDP}) \sum_{k=1}^{\text{NDP}} |P_{i,\text{lit}}^s - P_{i,\text{calcd}}^s / P_{i,\text{lit}}^s|$, where $P_{i,\text{lit}}^s$ is the literature vapor pressure and $P_{i,\text{calcd}}^s$ is the calculated vapor pressure using the determined Antoine constants.

Table 4. Experimental Saturated Vapor Pressures (P_i^s) of Pure Components Used in This Work at Various Temperatures (T)

Acetone		Ethanol		1-Butanol		2-Ethyl-1-hexanol	
P_i^s (kPa)	T (K)	P_i^s (kPa)	T (K)	P_i^s (kPa)	T (K)	P_i^s (kPa)	T (K)
40.00 ^a	304.35	40.00 ^a	329.50	40.00 ^a	366.69	15.40 ^a	401.29
53.33 ^a	311.55	53.33 ^a	335.90	53.33 ^a	373.67	26.66 ^a	415.59
66.66 ^a	317.43	66.66 ^a	341.14	66.66 ^a	379.46	40.00 ^a	427.24
79.99 ^a	322.56	79.99 ^a	345.45	79.99 ^a	384.30	53.33 ^a	435.96
93.32 ^a	326.96	93.32 ^a	349.25	93.32 ^a	388.50	66.66 ^a	443.14
100.8 ^b	329.07	100.5 ^b	351.20	100.3 ^b	390.53	79.99 ^a	449.22
						93.32 ^a	454.56
						101.2 ^b	457.44

^aStandard uncertainties are $u(P_i^s) = 0.05$ kPa and $u(T) = 0.05$ K. ^bStandard uncertainties are $u(P_i^s) = 0.25$ kPa and $u(T) = 0.06$ K.

tems are unattractive for industrial-scale processes because clogging and fouling of the membranes can occur.^{22,24,25} Adsorption has the lowest energy requirements of the above techniques but has a low capacity and selectivity and also results in fouling.^{19,22,24,25} Gas stripping is a simple process for purification but can result in the insufficient recovery of solvents.^{12,24,25} Distillation is a well-studied technology for purification and is easy to scale-up for industrial processes.^{15,20} Distillation techniques have been investigated for the separation of the products of ABE fermentation.^{8,26} However, distillation involves considerable energy consumption for the

purification of biobutanol. Distillation requires approximately 79.5 MJ/kg for the purification of 0.5 wt % biobutanol after fermentation to 99.9 wt %, which is more than two times the energy content of biobutanol (36 MJ/kg).⁸ Thus, a hybrid extraction–distillation process, i.e., a combination of liquid–liquid extraction and distillation, has been proposed for the large-scale production of biobutanol.^{15,19,24,27,28} In this process, first feed from the fermentation and extractant are fed into the extraction column located after the fermentor. In some cases, *Clostridium* bacteria are separated and recycled using a hollow-fiber ultrafilter before adding the extractant, in

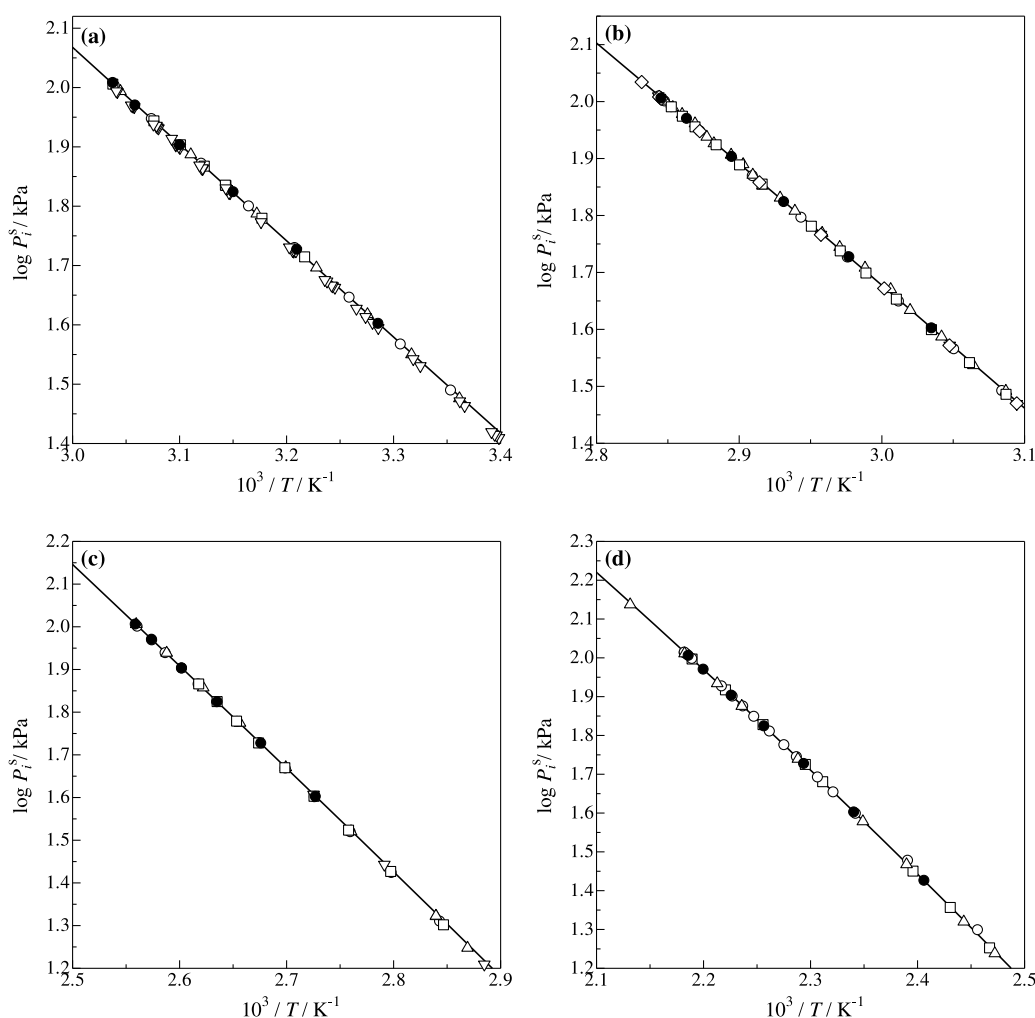


Figure 1. Experimental saturated pressures. (a) Acetone: ●, this work; ○, Ambrose et al.;⁷⁴ △, Boublik and Aim;⁷⁵ □, Brown and Smith;⁷⁶ ▽, Taylor.⁷⁷ (b) Ethanol: ●, this work; ○, Ambrose and Sprake;⁷⁸ △, Bourgeois et al.;⁷⁹ □, Figurski and Malanowski;⁸⁰ ▽, Kretschmer and Wiebe;⁸¹ ◇, Mund and Heim.⁸² (c) 1-Butanol: ●, this work; ○, Ambrose and Sprake;⁷⁸ △, Brown and Smith;⁸³ □, Kahlbaum;⁸⁴ ▽, Kemme and Kreps.⁸⁵ (d) 2-Ethyl-1-hexanol: ●, this work; ○, Laavi et al.;⁴⁹ △, Ambrose and Ghiassee;⁸⁶ □, Dykvy and Seprakova;⁸⁷ black line, Antoine equation.

order to increase the productivity of the ABE fermentation.^{29,30} After that, the raffinate phase contains water, traces of acetone, 1-butanol, and ethanol. The extract phase contained the extractant, acetone, 1-butanol, and ethanol. Then, the extract phase is fed into the distillation section for the recovery of the extractant, acetone, 1-butanol, and ethanol.

The selection of the extractant is crucial for the performance and economy of the process because the composition of the extract phase is directly affected by the extractant.¹⁹ Many extractant candidates, including branched alcohols (2-ethyl-1-hexanol^{24,31,32} and 2-butyl-1-octanol³³), 1-alkanols (1-heptanol³⁴ and 1-decanol^{18,35,36}), decane,³⁶ mesitylene,^{8,27} oleyl alcohol,^{13,37,38} 4-*n*-butyl-phenol,³⁹ *n*-hexyl acetate,^{15,19,28} bio-based lactates,¹⁷ soybean-derived biodiesel,⁴⁰ vegetable oil,³³ ionic liquids,^{41–44} and deep eutectic solvents,⁴⁵ have been considered. The properties of the extractants examined to date have been summarized in several reviews.^{6,7,18,21} 2-Ethyl-1-hexanol, a typical extractant, is less toxic and has moderately high distribution coefficients for the extraction of 1-butanol, and although 2-ethyl-1-hexanol is toxic to bacteria, it is considered less toxic than 1-octanol for instance.⁴⁶ Dalle and Adams¹⁸ have reported that 2-ethyl-1-hexanol is the most economic and environmentally friendly extractant and also has

a higher 1-butanol distribution coefficient than other extractants. Therefore, in the present study, 2-ethyl-1-hexanol was selected as the extractant for liquid–liquid extraction. As is obvious, liquid–liquid equilibrium data containing acetone, 1-butanol, ethanol, and 2-ethyl-1-hexanol have great importance for the design of the liquid–liquid extraction process. Then, the vapor–liquid equilibrium (VLE) data for binary and ternary mixtures containing 2-ethyl-1-hexanol will also be important to design the recovery of extractant and diluents by the distillation process. Herein, we focused on the distillation processes for the recovery of acetone, 1-butanol, ethanol, and 2-ethyl-1-hexanol in the extract phase in the hybrid extraction–distillation process coupled with 2-ethyl-1-hexanol as the extractant. Knowledge of the VLE data of the mixtures of these compounds is essential for the design of the distillation process. Predictions of the phase equilibria required for the process simulation or evaluation have been previously performed using the UNIFAC, the NRTL parameters built in Aspen Plus, and the COSMO-RS method.^{15,27} However, the group interaction parameters in the UNIFAC and NRTL parameters may not be available for some candidate extractants.²⁴ Therefore, obtaining accurate phase equilibrium data related to acetone, 1-butanol, ethanol, and the extractant

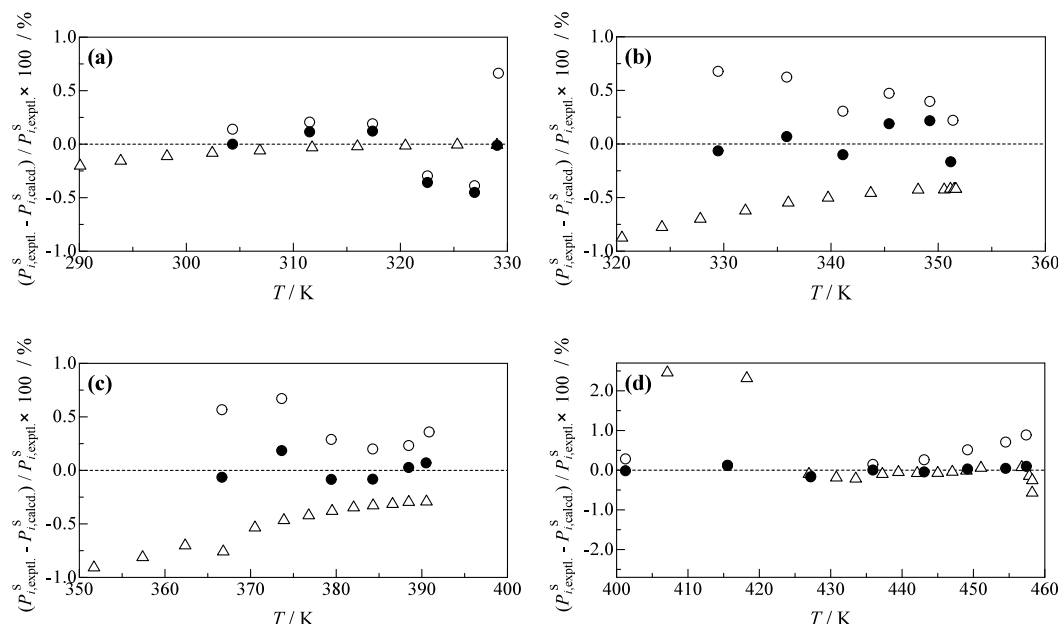


Figure 2. Relative deviations between the experimental and calculated saturated vapor pressures vs temperature for (a) acetone, (b) ethanol, (c) 1-butanol, and (d) 2-ethyl-1-hexanol. ●, deviations between the experimental and calculated vapor pressures using the determined Antoine constants; ○, deviations between the experimental and calculated vapor pressures using Antoine constants from the literature; △, deviations between the literature and the calculated vapor pressure using the determined Antoine constants.

Table 5. Experimental VLE Data of the Binary System Acetone (1) + Ethanol (2) at Pressure (P) of 101.3 kPa^a, Liquid Mole Fraction (x_1), Vapor Mole Fraction (y_1), and Temperature (T)^b

x_1	y_1	T (K)	γ_1	γ_2
0.022	0.076	349.98	1.768	0.999
0.024	0.081	349.89	1.730	0.998
0.044	0.138	348.76	1.694	1.000
0.055	0.166	348.22	1.646	1.000
0.085	0.237	346.76	1.576	1.004
0.100	0.269	346.05	1.556	1.006
0.126	0.318	344.98	1.518	1.009
0.137	0.339	344.49	1.511	1.011
0.164	0.383	343.43	1.471	1.017
0.182	0.414	342.79	1.463	1.014
0.200	0.441	342.13	1.443	1.018
0.242	0.492	340.91	1.386	1.027
0.269	0.522	340.12	1.355	1.036
0.302	0.556	339.25	1.319	1.047
0.350	0.595	338.13	1.265	1.075
0.395	0.629	337.11	1.225	1.105
0.446	0.665	336.26	1.180	1.130
0.487	0.692	335.58	1.148	1.158
0.526	0.716	334.86	1.127	1.190
0.558	0.731	334.51	1.097	1.229
0.615	0.762	333.59	1.070	1.301
0.644	0.776	333.27	1.052	1.340
0.690	0.799	332.66	1.031	1.420
0.738	0.827	332.05	1.018	1.490
0.774	0.848	331.46	1.015	1.559
0.811	0.869	331.07	1.005	1.639
0.918	0.940	329.94	0.999	1.802

^aCorrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. ^bStandard uncertainties, u , are $u(x_1) = 0.003$, $u(y_1) = 0.003$, and $u(T) = 0.06$ K.

Table 6. Experimental VLE Data of the Binary System Acetone (1) + 1-Butanol (2) at Pressure (P) of 101.3 kPa^a, Liquid Mole Fraction (x_1), Vapor Mole Fraction (y_1), and Temperature (T)^b

x_1	y_1	T (K)	γ_1	γ_2
0.113	0.569	371.71	1.416	1.001
0.198	0.734	362.30	1.343	1.011
0.249	0.794	357.79	1.312	1.019
0.369	0.874	349.47	1.239	1.094
0.451	0.902	345.44	1.181	1.191
0.614	0.943	339.47	1.095	1.326
0.783	0.971	334.54	1.038	1.535
0.908	0.989	331.45	1.009	1.634

^aCorrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. ^bStandard uncertainties, u , are $u(x_1) = 0.003$, $u(y_1) = 0.003$, and $u(T) = 0.06$ K.

and accurate parameters from activity coefficient models determined using these compounds are very important. Some experimental VLE data for mixtures containing 2-ethyl-1-hexanol can be available in the literature.^{47–53} However, these data are not sufficient. Thus, in the present work, we measured the experimental VLE of mixtures containing acetone, ethanol, 1-butanol, and 2-ethyl-1-hexanol. The VLE for two ternary systems was studied in the present work, i.e., acetone + ethanol + 1-butanol and ethanol + 1-butanol + 2-ethyl-1-hexanol, and the six constituent binary systems at 101.3 kPa. The experimental constituent binary VLE data were correlated by the Wilson,⁵⁴ NRTL,⁵⁵ and UNIQUAC⁵⁶ models. The VLE data for the two ternary systems were predicted from binary parameters in three models obtained from the constituent binary VLE data, and the predicted ternary VLE data were compared with the experimental results. On the other hand, the evaluation of the prediction accuracies of VLE by the group contribution method is also important for its application in process simulators. Two modifications of the

Table 7. Experimental VLE Data of the Binary System Ethanol (1) + 1-Butanol (2) at Pressure (P) of 101.3 kPa^a, Liquid Mole Fraction (x_1), Vapor Mole Fraction (y_1), and Temperature (T)^b

x_1	y_1	T (K)	γ_1	γ_2
0.073	0.231	385.13	0.955	1.018
0.117	0.345	382.23	0.970	1.015
0.133	0.379	381.03	0.983	1.024
0.160	0.445	379.04	1.020	1.020
0.177	0.483	377.89	1.039	1.014
0.220	0.556	375.67	1.039	1.001
0.246	0.595	374.10	1.051	1.004
0.275	0.640	372.49	1.068	0.990
0.311	0.682	370.85	1.064	0.985
0.337	0.707	369.93	1.053	0.978
0.390	0.748	367.66	1.041	1.005
0.444	0.793	365.76	1.039	0.980
0.473	0.811	364.63	1.039	0.990
0.503	0.824	363.82	1.021	1.014
0.530	0.838	362.87	1.022	1.026
0.553	0.851	362.25	1.017	1.016
0.623	0.881	360.35	1.002	1.050
0.677	0.904	358.62	1.010	1.062
0.716	0.919	357.90	0.997	1.061
0.762	0.936	356.76	0.996	1.056
0.821	0.954	355.21	1.000	1.080
0.844	0.959	354.51	1.006	1.123
0.878	0.971	354.05	0.995	1.064
0.928	0.982	352.92	0.996	1.170

^aCorrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. ^bStandard uncertainties, u , are $u(x_1) = 0.003$, $u(y_1) = 0.003$, and $u(T) = 0.06$ K.

Table 8. Experimental VLE Data of the Binary System Acetone (1) + 2-Ethyl-1-hexanol (2) at Pressure (P) of 101.3 kPa^a, Liquid Mole Fraction (x_1), Vapor Mole Fraction (y_1), and Temperature (T)^b

x_1	y_1	T (K)	γ_1	γ_2
0.278	0.978	367.04	1.120	0.984
0.301	0.981	364.64	1.111	0.988
0.327	0.984	361.51	1.116	1.026
0.377	0.988	356.96	1.104	1.033
0.403	0.990	354.92	1.098	1.045
0.475	0.993	349.67	1.088	1.110
0.530	0.995	346.49	1.076	1.156
0.587	0.996	343.46	1.067	1.204
0.614	0.996	342.58	1.049	1.226
0.653	0.997	340.82	1.043	1.234
0.683	0.998	339.71	1.034	1.250
0.726	0.998	338.25	1.018	1.279
0.765	0.998	336.92	1.010	1.383
0.821	0.999	334.87	1.006	1.478
0.849	0.999	334.00	1.002	1.552
0.881	0.999	332.99	0.999	1.685
0.907	0.999	332.17	0.997	1.724
0.937	1.000	331.47	0.989	1.765

^aCorrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. ^bStandard uncertainties, u , are $u(x_1) = 0.003$, $u(y_1) = 0.003$, and $u(T) = 0.06$ K.

original UNIFAC, i.e., modified UNIFAC (Dortmund),^{57–61} modified UNIFAC (Lyngby),⁶² and ASOG,^{63–68} have been

Table 9. Experimental VLE Data of the Binary System Ethanol (1) + 2-Ethyl-1-hexanol (2) at Pressure (P) of 101.3 kPa^a, Liquid Mole Fraction (x_1), Vapor Mole Fraction (y_1), and Temperature (T)^b

x_1	y_1	T (K)	γ_1	γ_2
0.172	0.905	395.71	1.140	0.953
0.254	0.945	385.03	1.124	0.977
0.290	0.955	381.40	1.119	0.983
0.322	0.964	378.45	1.123	0.953
0.419	0.977	371.52	1.107	0.983
0.459	0.981	369.62	1.084	0.953
0.510	0.985	367.02	1.073	0.966
0.567	0.988	364.51	1.060	0.971
0.586	0.989	363.79	1.053	0.975
0.655	0.992	361.16	1.042	1.044
0.713	0.994	359.45	1.021	1.082
0.779	0.995	357.31	1.015	1.122
0.896	0.998	354.64	0.981	1.265
0.911	0.998	354.21	0.981	1.297

^aCorrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. ^bStandard uncertainties, u , are $u(x_1) = 0.003$, $u(y_1) = 0.003$, and $u(T) = 0.06$ K.

Table 10. Experimental VLE Data of the Binary System 1-Butanol (1) + 2-Ethyl-1-hexanol (2) at Pressure (P) of 101.3 kPa^a, Liquid Mole Fraction (x_1), Vapor Mole Fraction (y_1), and Temperature (T)^b

x_1	y_1	T (K)	γ_1	γ_2
0.078	0.395	442.78	1.050	1.007
0.156	0.596	432.70	1.037	1.011
0.198	0.665	428.42	1.029	1.016
0.229	0.708	425.58	1.026	1.014
0.267	0.752	422.39	1.022	1.010
0.326	0.803	418.09	1.017	1.013
0.401	0.855	413.16	1.020	1.008
0.444	0.877	411.02	1.010	0.997
0.540	0.917	406.27	1.009	0.974
0.647	0.947	401.71	1.008	0.977
0.720	0.963	398.97	1.009	0.958
0.759	0.970	397.70	1.006	0.965
0.809	0.977	396.20	1.001	0.961
0.893	0.989	393.82	0.996	0.958

^aCorrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. ^bStandard uncertainties, u , are $u(x_1) = 0.003$, $u(y_1) = 0.003$, and $u(T) = 0.06$ K.

widely used for practical uses. Our group has tested the modified UNIFAC (Dortmund) and ASOG for the prediction of VLE of binary and ternary systems.^{69,70} As a continuation from our previous work, we also investigated these two group contribution models for the prediction of VLE of these binary and ternary systems.

2. EXPERIMENTAL SECTION

2.1. Materials. The chemicals used in this work are listed in Table 1. The purity of all chemicals was verified by gas chromatography (GC-1700, Shimadzu Co., Ltd., Kyoto, Japan) equipped with a flame ionization detector (FID). The purity of other except acetone was also checked by determining the water content using a Karl Fischer moisture meter (CA-200, Mitsubishi Chemical Co., Ltd., Tokyo, Japan).

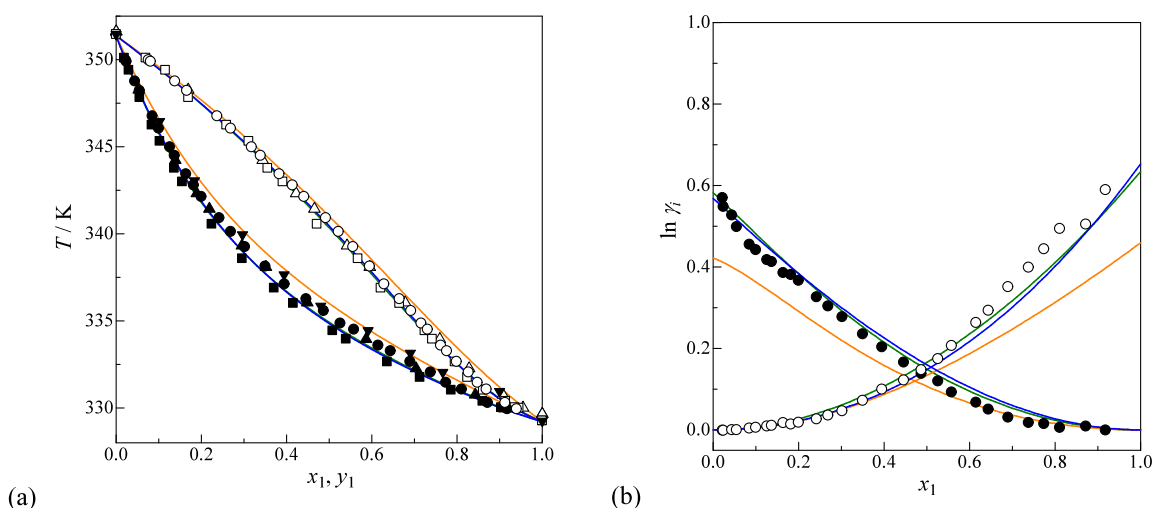


Figure 3. Experimental VLE for the system of acetone (1) + ethanol (2) at 101.3 kPa. (a) Temperature–composition relationships: closed symbol, $T-x_1$; open symbol, $T-y_1$; ● and ○, this work; ▲ and △, Ciprian and Vlaic;⁸⁹ ■ and □, Zhou and Han;⁹⁰ ▼, Adi et al.⁹¹ (b) Activity coefficient–composition relationships: ●, experimental $\ln \gamma_1-x_1$; ○, experimental $\ln \gamma_2-x_1$. blue line, NRTL; green line, modified UNIFAC (Dortmund); red line, ASOG.

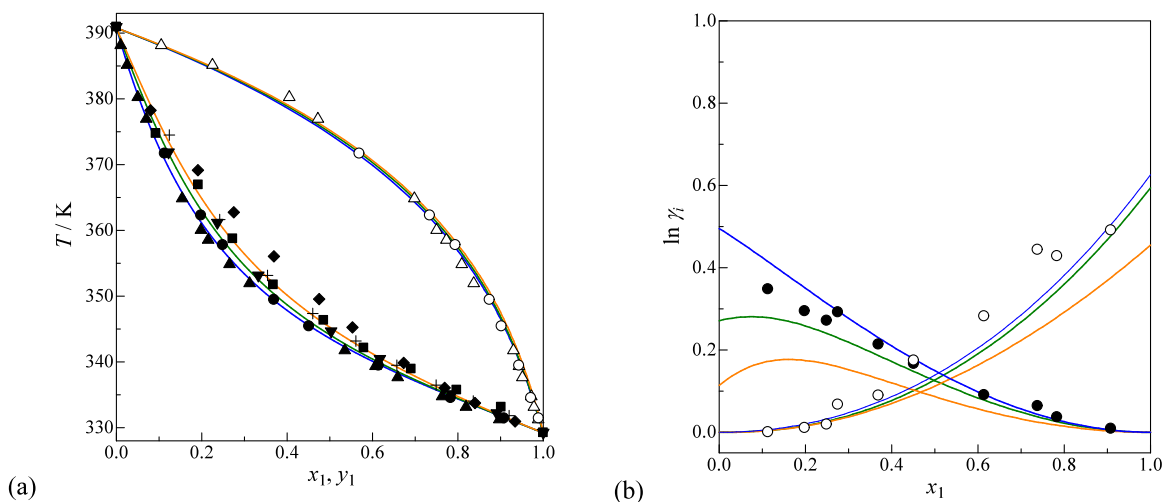


Figure 4. Experimental VLE for the system of acetone (1) + 1-butanol (2) at 101.3 kPa. (a) Temperature–composition relationships: closed symbol, $T-x_1$; open symbol, $T-y_1$; ● and ○, this work; ▲ and △, Michalski et al.;⁹² ■, Adi et al.;⁹¹ ▼, Ling et al.;⁹³ +, Rhim et al.;⁹⁴ +, Ernst et al.⁹⁵ (b) Activity coefficient–composition relationships: ●, experimental $\ln \gamma_1-x_1$; ○, experimental $\ln \gamma_2-x_1$. blue line, NRTL; green line, modified UNIFAC (Dortmund); orange line, ASOG.

The normal boiling points (T_b) and densities (ρ) at 298.15 K were also measured. T_b values were determined using a modified Rose–Williams equilibrium still (details are given in the next section). In the measurements of T_b , the experimental boiling point T_{exptl} was converted to T_b using eq 1:⁷¹

$$T_b = \frac{1}{\frac{1}{T_{\text{exptl}} + C_i} + \frac{\ln P_{\text{exptl}} - \ln 101.3}{2.303 B_i}} - C_i \quad (1)$$

where T_{exptl} is the experimental boiling point at the actual atmospheric pressure P_{exptl} ; and B_i and C_i are the Antoine constants. Descriptions for the Antoine constants are presented in Sections 2.2 and 3.1. The values for ρ at 298.15 K were measured by using a precision digital oscillating U-tube densimeter (DMA 4500, Anton Paar GmbH, Graz, Austria). The experimental T_b and ρ values at 298.15 K are listed in Table 2, along with the literature values.⁷² The standard uncertainties of T_b and ρ were estimated to be $u(T_b) = 0.06$ K

and $u(\rho) = 0.11$ kg/m³, respectively. Evaluation procedures for these standard uncertainties are described in the Supporting Information.

2.2. Experimental Apparatus and Procedure. We used a modified Rose–Williams equilibrium still for VLE measurements. The equilibrium still and experimental procedure have been described in detail in our previous work.⁷¹ The equilibrium temperature was measured with a calibrated 100 Ω platinum resistance thermometer. It took about (30 to 60) min to attain the equilibrium for binaries and (1 to 1.5) h for ternaries. The equilibrium still was open to the atmosphere through silica gel in order to absorb moisture in the air. The atmospheric pressure was determined with a Fortin barometer. The atmospheric pressure depended on the day, and it oscillated within 0.25 kPa during a measurement. Therefore, the experimental boiling points T were corrected to 101.3 kPa using eq 2:⁷³

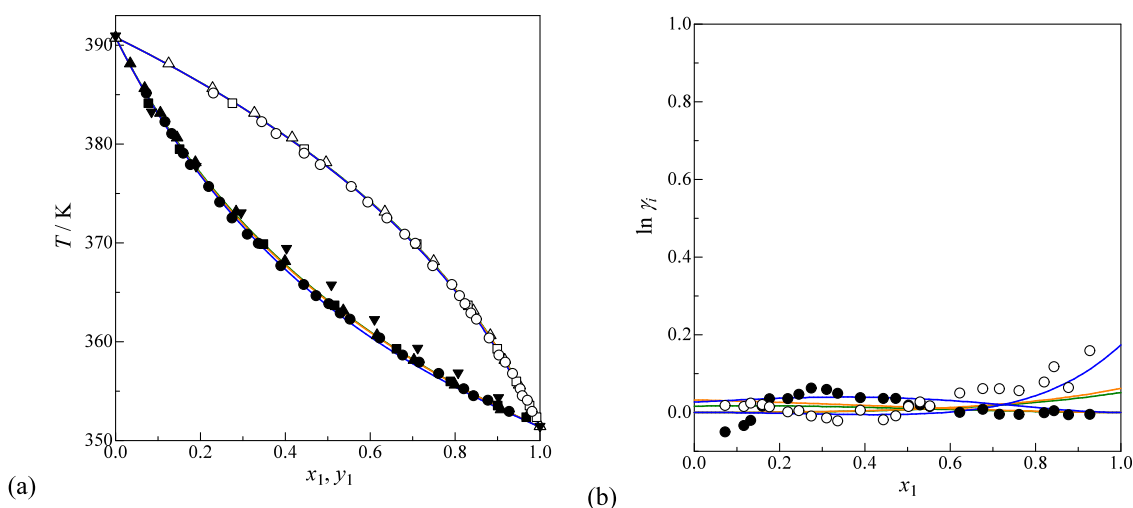


Figure 5. Experimental VLE for the system of ethanol (1) + 1-butanol (2) at 101.3 kPa. (a) Temperature–composition relationships: closed symbol, T – x_1 ; open symbol, T – y_1 ; ● and ○, this work; ▲ and △, Gay;⁹⁶ ■ and □, Kharin et al.;⁹⁷ ▼, Adi et al.⁹¹ (b) Activity coefficient–composition relationships: ●, experimental $\ln \gamma_1$ – x_1 ; ○, experimental $\ln \gamma_2$ – x_1 . blue line, NRTL; green line, modified UNIFAC (Dortmund); orange line, ASOG.

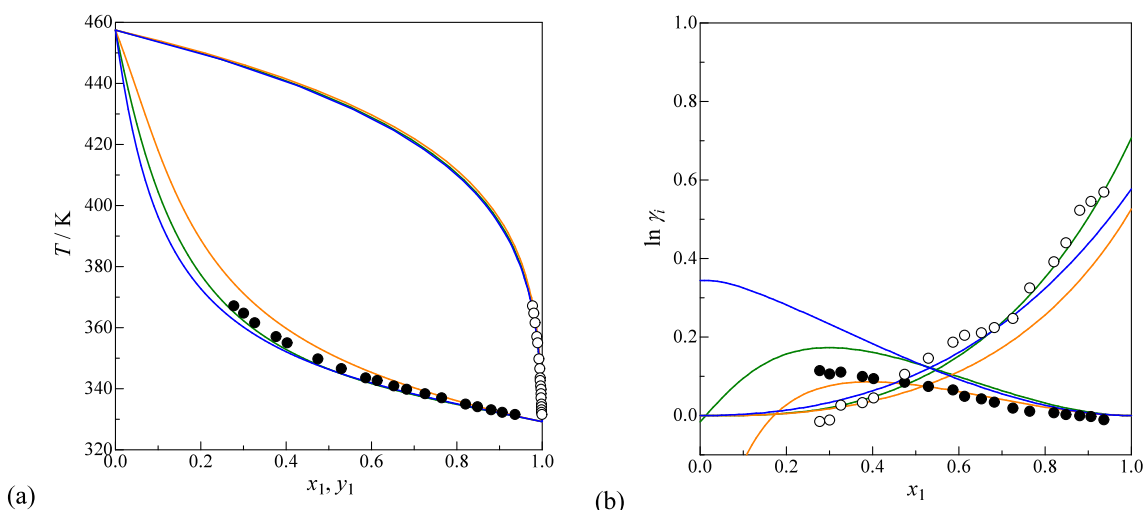


Figure 6. Experimental VLE for the system of acetone (1) + 2-ethyl-1-hexanol (2) at 101.3 kPa. (a) Temperature–composition relationships: closed symbol, T – x_1 ; open symbol, T – y_1 ; ● and ○, this work. (b) Activity coefficient–composition relationships: ●, experimental $\ln \gamma_1$ – x_1 ; ○, experimental $\ln \gamma_2$ – x_1 . blue line, NRTL; green line, modified UNIFAC (Dortmund); orange line, ASOG.

$$T = T_{\text{exptl}} + \frac{1}{2.303 \sum_{i=1}^{\text{NC}} B_i x_i / (T_{\text{exptl}} + C_i)^2} \frac{101.3 - P_{\text{exptl}}}{P_{\text{exptl}}} \quad (2)$$

where NC is the number of components; T_{exptl} is the experimental boiling point at the actual atmospheric pressure P_{exptl} ; B_i and C_i are the Antoine constants of component i in eq 3, as listed in Table 3, and x_i is the mole fraction in the liquid phase. The standard uncertainty of T , $u(T)$, is described later because $u(x_i)$ should be used in the estimation.

2.3. Analysis. Vapor and liquid samples were analyzed by using a gas chromatograph (GC-1700, Shimadzu Co., Ltd., Kyoto, Japan) equipped with an FID. ULBON HR-20 M (50 × 0.25 mm I.D., 0.50 μm film thickness; Shinwa Chemical Industries Ltd., Kyoto, Japan) was used for column packing. Injections (0.2 μL) were performed in split mode at a split ratio of 85/1. Helium was used as the carrier gas at a flow rate of 2.0 mL min^{−1}. The detector temperature was maintained at 523 K. Compositions were determined by the relative area

method with a calibration curve. The standard uncertainties of the composition for liquid and vapor phases, i.e., $u(x_i)$ and $u(y_i)$, respectively, were estimated to be $u(x_i) = u(y_i) = 0.003$ for binaries and $u(x_i) = u(y_i) = 0.004$ for ternaries. Considering the values of $u(x_i)$, the standard uncertainties for the boiling point were estimated to be $u(T) = 0.06$ and 0.11 K for binaries and ternaries, respectively. Detailed descriptions for these uncertainty evaluations can be found in the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Saturated Vapor Pressures. The Antoine constants A_i , B_i , and C_i in eq 3 were determined from the experimental vapor pressure data through regression by minimizing the objective function (F_{obj}) given by eq 4:

$$\log P_i^s = A_i - \frac{B_i}{T + C_i} \quad (3)$$

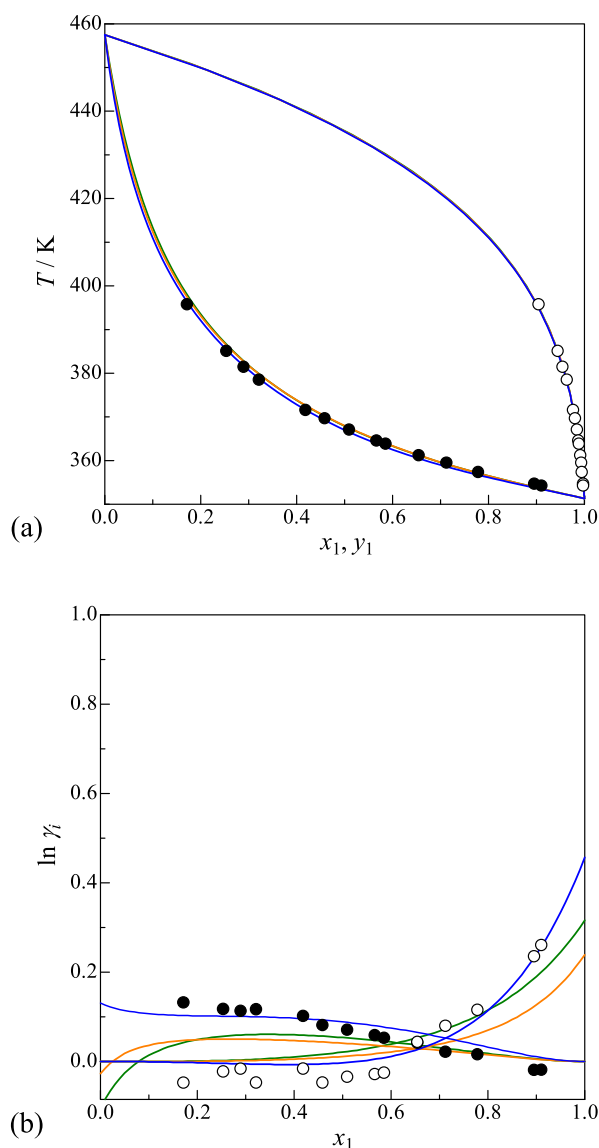


Figure 7. Experimental VLE for the system of ethanol (1) + 2-ethyl-1-hexanol (2) at 101.3 kPa. (a) Temperature–composition relationships: closed symbol, $T-x_1$; open symbol, $T-y_1$; ● and ○, this work; ▲ and △, Ghanadzadeh et al.⁵³ (b) Activity coefficient–composition relationships: ●, experimental $\ln \gamma_1-x_1$; ○, experimental $\ln \gamma_2-x_1$. blue line, NRTL; green line, modified UNIFAC (Dortmund); orange line, ASOG.

$$F_{\text{obj}} = \sum_{k=1}^{\text{NDP}} \left(\frac{P_{i,\text{exptl}}^s - P_{i,\text{calcd}}^s}{P_{i,\text{exptl}}^s} \right)^2 \quad (4)$$

where NDP is the number of experimental vapor pressures, and $P_{i,\text{exptl}}^s$ and $P_{i,\text{calcd}}^s$ are the experimental and calculated vapor pressures, respectively. Determined Antoine constants are listed in Table 3. Table 4 shows the experimental saturated vapor pressures (P_i^s) for acetone, ethanol, 1-butanol, and 2-ethyl-1-hexanol. Figure 1 shows a comparison of the experimental saturated vapor pressures with the literature values^{49,74–87} by a plot of the logarithm of the saturated vapor pressure ($\log P_i^s$) as a function of the inverse of absolute temperature ($10^3/T$). The experimental values for these four compounds agreed with the literature data. The average relative deviations between the experimental values and calculated values using the literature values for the Antoine

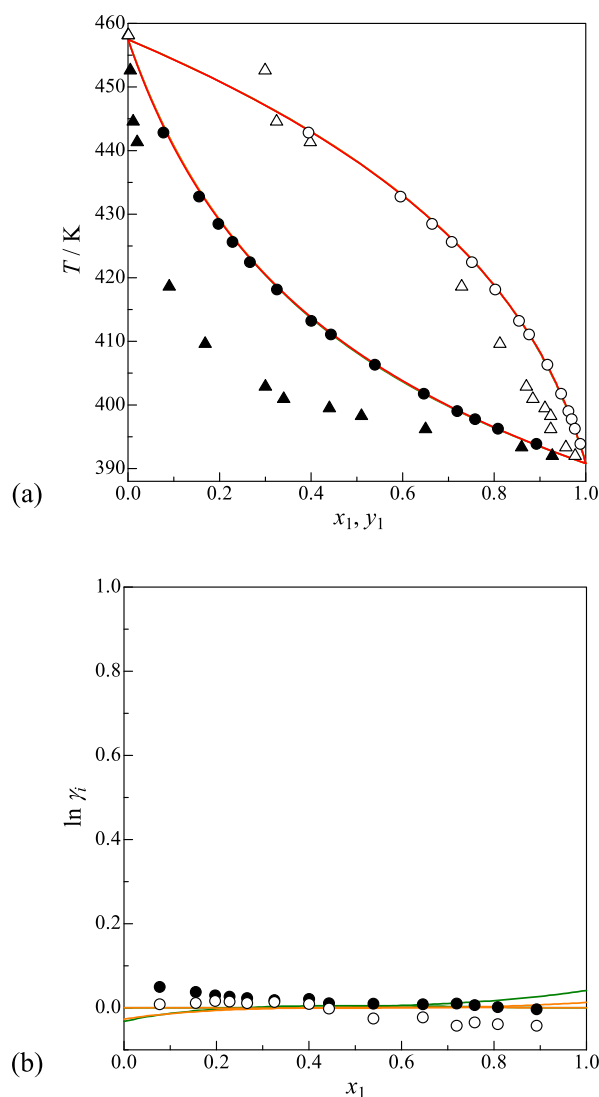


Figure 8. Experimental VLE for the system of 1-butanol (1) + 2-ethyl-1-hexanol (2) at 101.3 kPa. (a) Temperature–composition relationships: closed symbol, $T-x_1$; open symbol, $T-y_1$; ● and ○, this work; ▲ and △, Ghanadzadeh et al.⁵³ (b) Activity coefficient–composition relationships: ●, experimental $\ln \gamma_1-x_1$; ○, experimental $\ln \gamma_2-x_1$. red line, Ideal solution; green line, modified UNIFAC (Dortmund); orange line, ASOG.

Table 11. Results of Point-To-Point Test^a for the Constituent Binary Systems at 101.3 kPa

System	$ \Delta y_1 _{\text{av}}$	
Acetone (1) + Ethanol (2)	0.006	+
Acetone (1) + 1-Butanol (2)	0.004	+
Ethanol (1) + 1-Butanol (2)	0.005	+
Acetone (1) + 2-Ethyl-1-hexanol (2)	0.001	+
Ethanol (1) + 2-Ethyl-1-hexanol (2)	0.001	+
1-Butanol (1) + 2-Ethyl-1-hexanol (2)	0.002	+

^aReference 99.

constants,^{71,88} $|(P_{i,\text{exptl}}^s - P_{i,\text{calcd}}^s)/P_{i,\text{exptl}}^s| \times 100$ (%), were within 0.44%. The average relative deviations between the literature^{49,74,78} and calculated values using the determined Antoine constants, $|(P_{i,\text{lit}}^s - P_{i,\text{calcd}}^s)/P_{i,\text{lit}}^s| \times 100$ (%), were within 0.95%. Figure 2 shows the relative deviations between

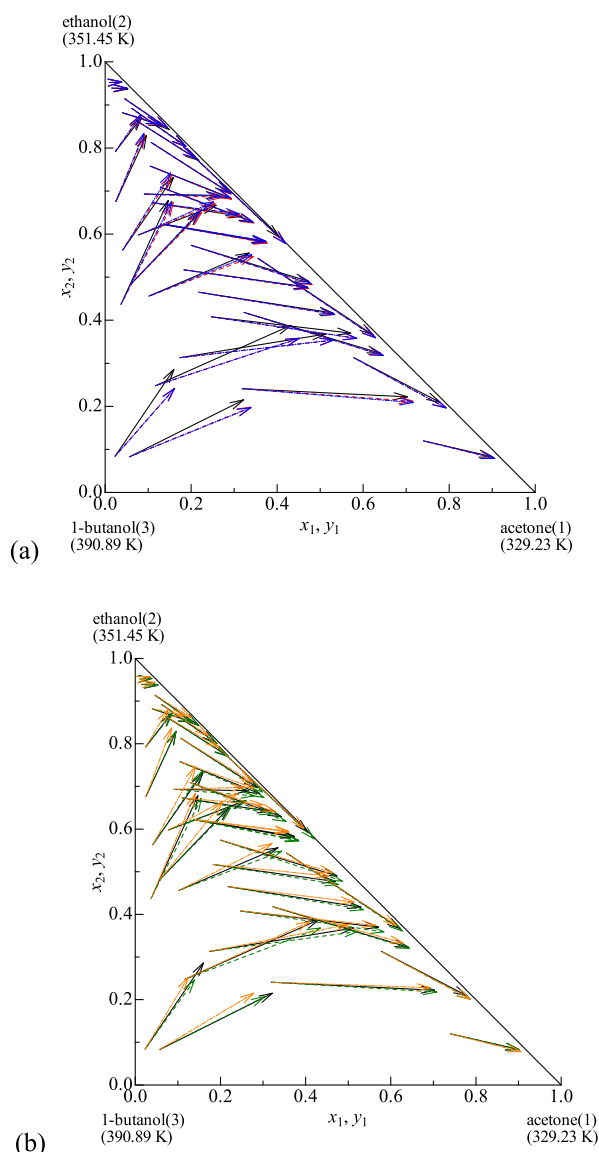


Figure 9. VLE tie lines (tails of arrows represent the liquid-phase mole fractions x_1 and x_2 , and the heads of arrows represent the vapor-phase mole fractions y_1 and y_2) of the system acetone (1) + ethanol (2) + 1-butanol (3) at 101.3 kPa. (a) Comparison between the experimental and predicted VLE parameters using the Wilson and NRTL model. (b) Comparison between the experimental and predicted VLE parameters using the modified UNIFAC (Dortmund) and ASOG. black arrows, this work; red arrows, Wilson; blue arrows, NRTL; green arrows, modified UNIFAC (Dortmund); orange arrows, ASOG.

the experimental and calculated values for P_i^s , $(P_{i,\text{exptl}}^s - P_{i,\text{calcd}}^s)/P_{i,\text{exptl}}^s \times 100$ (%), as a function of the temperature T (K).

3.2. Binary Systems. We measured the VLE data for six constituent binary systems, i.e., acetone + ethanol, acetone + 1-butanol, ethanol + 1-butanol, acetone + 2-ethyl-1-hexanol, ethanol + 2-ethyl-1-hexanol, and 1-butanol + 2-ethyl-1-hexanol, at 101.3 kPa. The experimental VLE data and the values for the activity coefficients of component i (γ_i) for these systems are summarized in Tables 5–10. The values for γ_i were calculated using eq 5 assuming ideal gas behavior:

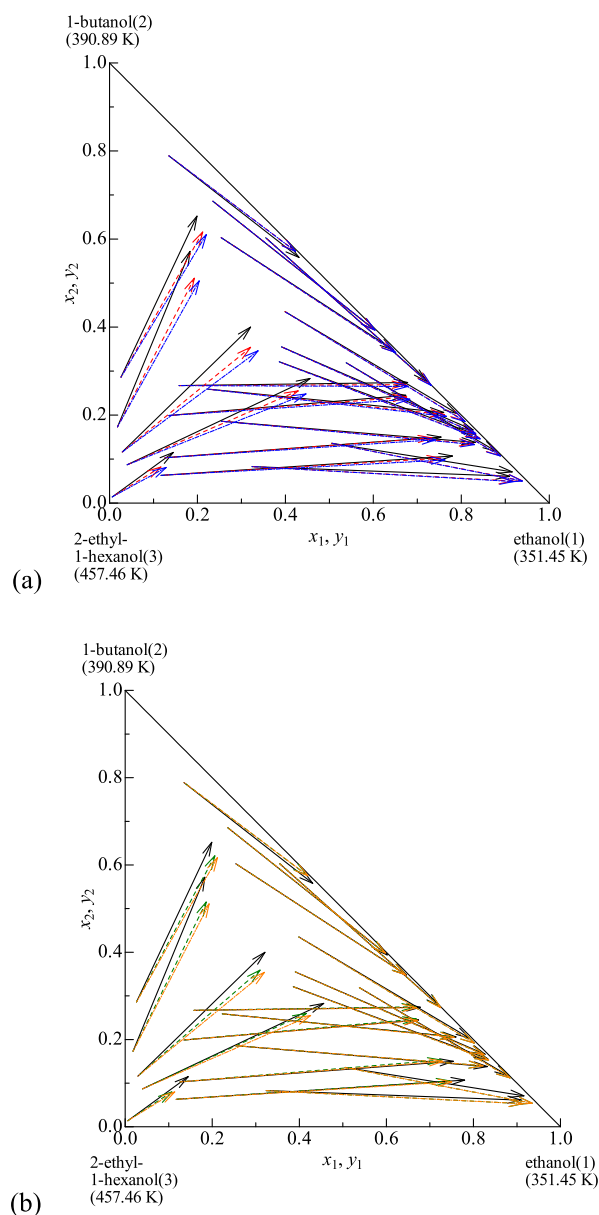


Figure 10. VLE tie lines (tails of arrows represent the liquid-phase mole fractions x_1 and x_2 , and heads of arrows represent the vapor-phase mole fractions y_1 and y_2) of the system ethanol (1) + 1-butanol (2) + 2-ethyl-1-hexanol (3) at 101.3 kPa. (a) Comparison between the experimental and predicted VLE parameters using the Wilson and NRTL model. (b) Comparison between the experimental and predicted VLE parameters using the modified UNIFAC (Dortmund) and ASOG. black arrows, this work; red arrows, Wilson; blue arrows, NRTL; green arrows, modified UNIFAC (Dortmund); orange arrows, ASOG.

$$\gamma_i = \frac{P_{y_i}}{P_i^s x_i} \quad (5)$$

P_i^s in eq 5 was calculated from the Antoine equation using the Antoine constants listed in Table 3. Diagrams of the experimental temperatures and γ_i versus composition are illustrated in Figures 3–8. For the systems acetone + ethanol, acetone + 1-butanol, and ethanol + 1-butanol, Figures 3(a), 4(a), and 5(a) compare the experimental VLE data with the literature ones^{89–97} including only T – x_1 diagrams reported. From these figures, it can be observed that the experimental

Table 12. Experimental VLE Data for the Ternary System Acetone (1) + Ethanol (2) + 1-Butanol (3) at a Pressure (P) of 101.3 kPa^a, Liquid Mole Fraction (x_1 , x_2), Vapor Mole Fraction (y_1 , y_2), and Temperature (T)^b

x_1	x_2	y_1	y_2	T (K)	γ_1	γ_2	γ_3
0.091	0.693	0.266	0.690	350.51	1.484	1.030	1.059
0.355	0.544	0.623	0.366	339.64	1.244	1.088	0.967
0.323	0.418	0.641	0.323	342.64	1.280	1.101	1.060
0.183	0.517	0.470	0.477	348.33	1.392	1.041	1.019
0.137	0.623	0.373	0.583	349.06	1.444	1.026	1.021
0.110	0.673	0.318	0.641	349.70	1.504	1.018	1.021
0.105	0.758	0.293	0.683	348.52	1.504	1.009	1.001
0.040	0.882	0.133	0.851	350.50	1.689	0.999	1.067
0.072	0.864	0.218	0.771	348.45	1.635	1.002	0.985
0.062	0.892	0.188	0.804	348.75	1.623	1.000	0.983
0.046	0.914	0.149	0.843	349.41	1.700	0.997	1.095
0.200	0.574	0.474	0.490	346.32	1.365	1.045	1.013
0.129	0.709	0.343	0.629	347.86	1.461	1.020	1.020
0.218	0.465	0.530	0.417	347.51	1.350	1.046	1.003
0.319	0.241	0.703	0.222	346.87	1.248	1.103	1.055
0.182	0.803	0.405	0.593	343.04	1.418	1.035	0.997
0.138	0.622	0.374	0.581	348.82	1.447	1.034	1.056
0.078	0.598	0.258	0.667	353.90	1.521	1.010	1.027
0.174	0.313	0.513	0.368	354.11	1.347	1.056	1.020
0.102	0.456	0.335	0.556	355.64	1.436	1.033	1.011
0.059	0.483	0.220	0.650	358.64	1.498	1.017	1.017
0.041	0.563	0.159	0.732	357.92	1.589	1.010	1.018
0.247	0.408	0.571	0.370	347.30	1.292	1.067	1.036
0.025	0.676	0.096	0.829	356.44	1.641	1.007	0.991
0.015	0.939	0.053	0.937	351.38	1.749	0.997	1.085
0.011	0.952	0.039	0.953	351.46	1.751	0.997	1.075
0.037	0.437	0.147	0.678	362.35	1.439	1.021	1.013
0.024	0.792	0.085	0.871	353.78	1.634	1.001	1.067
0.740	0.120	0.901	0.083	334.09	1.033	1.424	1.360
0.057	0.083	0.322	0.215	375.67	1.440	1.065	0.947
0.578	0.313	0.779	0.209	335.82	1.081	1.274	1.194
0.107	0.813	0.287	0.697	347.11	1.508	1.017	1.223
0.117	0.248	0.432	0.388	361.09	1.385	1.079	0.911
0.023	0.084	0.160	0.286	380.76	1.562	1.181	0.896

^aCorrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. ^bStandard uncertainties, u , are $u(x_1) = u(x_2) = 0.004$, $u(y_1) = u(y_2) = 0.004$, and $u(T) = 0.11$ K.

data agreed well with the literature data^{89,90,92,96,97} which are thermodynamically consistent with the point-to-point test of van Ness et al.⁹⁸ with modifications by Fredenslund et al.⁹⁹ However, differences occurred between the experimental and other literature values in the T – x_1 diagram. In the system 1-butanol + 2-ethyl-1-hexanol, Ghanadzadeh et al. have reported the experimental VLE data at 101.3 kPa,⁵³ and their data are also plotted in Figure 8(a). As is clear from the figure, the VLE behavior in this work differs significantly from our experimental one. These literature data are considered to be doubtful data by the Dortmund Data Bank 2023.¹⁰⁰ The system acetone + 1-butanol contains fewer data points than other systems because we confirmed the agreement between the experimental and literature values. The systems acetone or ethanol + 2-ethyl-1-hexanol have large boiling point differences between pure components, about 128 and 106 K, respectively, and the slope of the T – x_1 curve changes rapidly. Therefore, we could not obtain reliable experimental data at a mole fraction close to pure 2-ethyl-1-hexanol due to large uncertainties of temperature T . The system 1-butanol + 2-ethyl-1-hexanol showed nearly ideal solution behavior from the values of γ_i in Table 10 and Figure 8. This system also contains fewer data

points due to the ideal solution behavior. The system 1-butanol + 2-ethyl-1-hexanol showed almost a constant value of γ_i (Table 10 and Figure 8), which follows the ideal solution. Therefore, the authors did not take many data points.

The experimental binary VLE data were tested for thermodynamic consistency using the aforementioned point-to-point test. The results of the thermodynamic consistency evaluation are shown in Table 11. According to this test, the experimental VLE data were thermodynamically consistent if the mean absolute deviation between the experimental and calculated vapor phase mole fractions, $|\Delta y_1|_{\text{av}}$, was less than 0.01. All of the experimental VLE data for the six binary systems were thermodynamically consistent.

3.3. Ternary Systems. Figures 9 and 10 and Tables 12 and 13 show the experimental VLE data for the ternary systems of acetone + ethanol + 1-butanol and ethanol + 1-butanol + 2-ethyl-1-hexanol at 101.3 kPa. The tails of the solid arrows in Figures 9 and 10 denote the experimental liquid compositions, and the heads of the arrows denote the experimental vapor compositions on the same tie line. As indicated in the figures, the two ternary systems did not form ternary azeotropic points. Adi et al.⁹¹ reported the VLE data of the system acetone +

Table 13. Experimental VLE Data for the Ternary System Ethanol (1) + 1-Butanol (2) + 2-Ethyl-1-hexanol (3) at Pressure (P) of 101.3 kPa^a, Liquid Mole Fraction (x_1, x_2), Vapor Mole Fraction (y_1, y_2), and Temperature (T)^b

x_1	x_2	y_1	y_2	T (K)	γ_1	γ_2	γ_3
0.399	0.435	0.791	0.203	368.55	1.043	1.093	1.059
0.391	0.355	0.827	0.166	369.93	1.060	1.035	0.752
0.223	0.259	0.760	0.206	382.66	1.111	1.070	0.965
0.135	0.199	0.674	0.245	394.92	1.108	1.066	1.040
0.040	0.087	0.456	0.283	422.26	1.173	1.186	0.896
0.029	0.116	0.321	0.400	424.72	1.069	1.171	0.899
0.324	0.083	0.910	0.061	376.19	1.135	1.266	0.977
0.505	0.136	0.916	0.071	365.39	1.069	1.396	1.252
0.126	0.103	0.754	0.150	399.09	1.172	1.094	0.895
0.006	0.014	0.145	0.115	449.22	1.290	1.453	0.956
0.135	0.789	0.431	0.558	381.81	1.070	0.982	2.213
0.538	0.319	0.880	0.117	363.42	1.035	1.066	0.806
0.386	0.321	0.833	0.156	370.14	1.074	1.067	1.014
0.158	0.267	0.678	0.274	389.54	1.124	1.073	0.900
0.235	0.686	0.602	0.394	375.53	1.059	1.016	1.045
0.355	0.603	0.722	0.276	369.82	1.024	1.018	1.307
0.254	0.603	0.646	0.348	375.16	1.064	1.035	0.881
0.254	0.186	0.832	0.138	380.78	1.136	1.071	0.859
0.117	0.063	0.780	0.107	402.59	1.179	1.136	0.860
0.026	0.286	0.199	0.652	413.55	0.994	1.078	0.888
0.018	0.173	0.183	0.572	424.50	0.987	1.130	0.840

^aCorrected value from the atmospheric pressure at measurement P_{exptl} with the standard uncertainty $u(P_{\text{exptl}}) = 0.25$ kPa. ^bStandard uncertainties, u , are $u(x_1) = u(x_2) = 0.004$, $u(y_1) = u(y_2) = 0.004$, and $u(T) = 0.11$ K.

ethanol + 1-butanol, but the vapor-phase mole fractions y_i are not reported in their VLE data. Therefore, we predicted the equilibrium temperatures T in the liquid-phase mole fractions x_i in the literature VLE from the binary parameters of the Wilson, NRTL, and UNIQUAC models listed in Table 15 and compared these equilibrium temperatures T with the literature ones because it was difficult to directly compare the experimental values with the literature ones in Figure 9. Details for the prediction are presented in Section 4.2. Average absolute deviations of the experimental and calculated T values, $|\Delta T|_{\text{av}}$, for the Wilson, NRTL, and UNIQUAC models were 6.09, 6.21, and 6.33 K, respectively, and large deviations were detected.

4. CORRELATION AND PREDICTION

4.1. Binary Systems. The experimental VLE data for the six constituent binary systems were correlated using Wilson,

Table 14. Molecular Size Parameter r_i and Molecular Area Parameter q_i of the UNIQUAC Model^a

Component	r_i	q_i
Acetone	2.5735	2.3360
Ethanol	2.1055	1.9720
1-Butanol	3.4543	3.0520
2-Ethyl-1-hexanol	6.1511	5.2080

^aReference 100.

NRTL, and UNIQUAC models. The liquid molar volume for the pure component v_i^L in the Wilson model was determined from the density measured at 298.15 K. The values of v_i^L are presented in Table 2. The nonrandomness parameter, α_{12} , in the NRTL model was set to 0.3 according to the recommendation of Renon and Prausnitz.⁵⁵ Molecular size and area values of pure component i in the UNIQUAC model,

i.e., r_i and q_i , respectively, were cited from the Dortmund Data Bank 2023.¹⁰⁰ These values are given in Table 14. Equation 6 was used to calculate F_{obj} for the regression of the binary parameters of the Wilson, NRTL, and UNIQUAC models, i.e., $\lambda_{ij} - \lambda_{ji}$, $g_{ij} - g_{ji}$, and $u_{ij} - u_{ji}$, respectively:

$$F_{\text{obj}} = \sum_{k=1}^{\text{NDP}} \left[\left(\frac{\gamma_{1,\text{exptl}} - \gamma_{1,\text{calcd}}}{\gamma_{1,\text{exptl}}} \right)_k^2 + \left(\frac{\gamma_{2,\text{exptl}} - \gamma_{2,\text{calcd}}}{\gamma_{2,\text{exptl}}} \right)_k^2 \right] \quad (6)$$

where $\gamma_{i,\text{exptl}}$ and $\gamma_{i,\text{calcd}}$ are the experimental and calculated activity coefficients of component i , respectively. The VLE of the system 1-butanol + 2-ethyl-1-hexanol was calculated as an ideal solution based on the γ_i for the system given in the previous section. The determined parameters for three models along with the average absolute deviations of the experimental and calculated T and y_1 values, $|\Delta T|_{\text{av}}$ and $|\Delta y_1|_{\text{av}}$, respectively, are provided in Tables 15 and 16. Three models gave good correlation accuracies for all the binary systems investigated in this work, giving the maximum values of deviations among these models $|\Delta T|_{\text{av}}$ and $|\Delta y_1|_{\text{av}}$ 0.14 K and 0.001, respectively. As an example of the correlation results of these models, the results of the NRTL model are graphically presented in Figures 3–8.

The modified UNIFAC (Dortmund) and ASOG were used to predict the VLE in the investigated constituent binary systems. In the modified UNIFAC (Dortmund), the van der Waals volume R_k and surface area Q_k of the functional groups and the group interaction parameters a_{nm} , b_{nm} , and c_{nm} were obtained from the Dortmund Data Bank 2023.¹⁰⁰ As for the ASOG, the group interaction parameters $m_{k/l}$ and $n_{k/l}$ were obtained from Tochigi et al.⁶⁴ $|\Delta T|_{\text{av}}$ and $|\Delta y_1|_{\text{av}}$ in both models are given in Tables 15 and 16, and the predicted results are also shown in Figures 3–8. In the systems acetone + ethanol and acetone + 1-butanol, $|\Delta T|_{\text{av}}$ and $|\Delta y_1|_{\text{av}}$ by the

Table 15. Determined Parameters and Deviations between the Experimental and Calculated Temperatures $|\Delta T|_{av}$ and Vapor-Phase Mole Fractions $|\Delta y|_{av}$ for the Wilson, NRTL, UNIQUAC, Modified UNIFAC (Dortmund), and ASOG for the Ternary System Acetone (1) + Ethanol (2) + 1-Butanol (3) and Its Constituent Binary Systems at 101.3 kPa

	Acetone (1) + Ethanol (2)	Acetone (1) + 1-Butanol (2)	Ethanol (1) + 1-Butanol (2)	Acetone (1) + Ethanol (2) + 1-Butanol (3)
Wilson Parameters				
$\lambda_{12} - \lambda_{11}$ (J mol ⁻¹)	80.287	1290.649	-506.118	—
$\lambda_{21} - \lambda_{22}$ (J mol ⁻¹)	1781.599	434.434	1791.088	—
Deviation				
$ \Delta T _{av}$ (K) ^a	0.27	0.47	0.23	0.30
$ \Delta y_1 _{av}$ ^b	0.007	0.006	0.006	0.005
$ \Delta y_2 _{av}$ ^b	—	—	—	0.007
NRTL Parameters				
$g_{12} - g_{22}$ (J mol ⁻¹)	1280.287	1198.552	4193.718	—
$g_{21} - g_{11}$ (J mol ⁻¹)	534.074	475.502	-2756.582	—
α_{12}	0.3	0.3	0.3	—
Deviation				
$ \Delta T _{av}$ (K) ^a	0.29	0.46	0.23	0.25
$ \Delta y_1 _{av}$ ^b	0.007	0.006	0.006	0.006
$ \Delta y_2 _{av}$ ^b	—	—	—	0.007
UNIQUAC Parameters				
$u_{12} - u_{22}$ (J mol ⁻¹)	1094.492	-101.484	1189.679	—
$u_{21} - u_{11}$ (J mol ⁻¹)	-177.347	792.818	-793.533	—
Deviation				
$ \Delta T _{av}$ (K) ^a	0.27	0.42	0.20	0.22
$ \Delta y_1 _{av}$ ^b	0.007	0.005	0.006	0.007
$ \Delta y_2 _{av}$ ^b	—	—	—	0.008
Modified UNIFAC (Dortmund)				
Deviation				
$ \Delta T _{av}$ (K) ^a	0.30	0.72	0.29	0.18
$ \Delta y_1 _{av}$ ^b	0.005	0.007	0.006	0.007
$ \Delta y_2 _{av}$ ^b	—	—	—	0.007
ASOG				
Deviation				
$ \Delta T _{av}$ (K) ^a	0.57	1.85	0.25	0.64
$ \Delta y_1 _{av}$ ^b	0.014	0.017	0.006	0.014
$ \Delta y_2 _{av}$ ^b	—	—	—	0.011

^a $|\Delta T|_{av} = \sum_{k=1}^{NDP} |T_{\text{exptl}} - T_{\text{calcd}}|_k / NDP$, where NDP is the number of data points. ^b $|\Delta y|_{av} = \sum_{k=1}^{NDP} |y_{\text{exptl}} - y_{\text{calcd}}|_k / NDP$.

ASOG were 0.57 and 0.014 and 1.85 and 0.017, respectively, and the prediction accuracies using the ASOG were slightly inferior to those using the modified UNIFAC (Dortmund). These results may be because the reproduction of the experimental values of the liquid-phase activity coefficients, i.e., $\ln \gamma_1$ and $\ln \gamma_2$ shown in Figures 3–8, was not sufficient compared to the other activity coefficient models.

4.2. Ternary Systems. We investigated the VLE predictions for two ternary systems using the Wilson, NRTL, UNIQUAC, modified UNIFAC (Dortmund), and ASOG models. First, the binary parameters, i.e., λ_{ij} – λ_{ji} , g_{ij} – g_{ji} , and u_{ij} – u_{ji} in the Wilson, NRTL, and UNIQUAC models, respectively, determined from the constituent binary systems were used to predict the VLE parameters. The modified UNIFAC (Dortmund) and ASOG were also investigated. The results are summarized in Tables 15 and 16. A comparison of the predicted results for the two ternary systems using the Wilson, NRTL, modified UNIFAC (Dortmund), and ASOG models with the experimental VLE data is shown in Figures 9 and 10. From the values of $|\Delta T|_{av}$, $|\Delta y_1|_{av}$, and $|\Delta y_2|_{av}$ in these tables, the prediction results from the Wilson, NRTL, and UNIQUAC models indicated that the binary parameters λ_{ij} – λ_{ji} , g_{ij} – g_{ji} , and u_{ij} – u_{ji} determined from the experimental VLE data of the binary constituent systems could reproduce the experimental

VLE data with almost good accuracy, although the UNIQUAC model gave slightly inferior results compared to the Wilson and NRTL models in the system ethanol + 1-butanol + 2-ethyl-1-hexanol, giving $|\Delta T|_{av}$ and $|\Delta y_2|_{av}$, 1.46 K and 0.024, respectively. It can be also concluded that the modified UNIFAC (Dortmund) and ASOG can also be used for the prediction of two ternary VLEs as well as with the Wilson, NRTL, and UNIQUAC models, from the maximum deviations in two ternary systems of $|\Delta T|_{av}$, $|\Delta y_1|_{av}$, and $|\Delta y_2|_{av}$, 0.73 K, 0.009, and 0.011, respectively, although the ASOG showed slightly poorer prediction accuracy in the system acetone + ethanol + 1-butanol with $|\Delta T|_{av} = 0.64$ K, $|\Delta y_1|_{av} = 0.014$, and $|\Delta y_2|_{av} = 0.011$.

5. CONCLUSIONS

The isobaric VLE data of two ternary systems, i.e., acetone + ethanol + 1-butanol and ethanol + 1-butanol + 2-ethyl-1-hexanol, and the six constituent binary systems were measured at 101.3 kPa. The experimental VLE data of the six binary systems were thermodynamically consistent, according to the point-to-point test. The experimental VLE data of these binary systems were well correlated by using the Wilson, NRTL, and UNIQUAC models. The experimental ternary VLE data were compared with the predicted data using the binary parameters

Table 16. Determined Parameters and Deviations between the Experimental and Calculated Temperatures $|\Delta T|_{av}$ and Vapor-Phase Mole Fractions $|\Delta y_i|_{av}$ for the Wilson, NRTL, UNIQUAC, Modified UNIFAC (Dortmund), and ASOG for the Ternary System Ethanol (1) + 1-Butanol (2) + 2-Ethyl-1-hexanol (3) and Its Constituent Binary Systems at 101.3 kPa

	Ethanol (1) + 1-Butanol (2)	Acetone (1) + 2-Ethyl-1-hexanol (2)	Ethanol (1) + 2-Ethyl-1-hexanol (2)	1-Butanol (1) + 2-Ethyl-1-hexanol (2)	Ethanol (1) + 1-Butanol (2) + 2-Ethyl-1-hexanol (3)
Wilson Parameters					
$\lambda_{12} - \lambda_{11}$ (J mol ⁻¹)	−506.118	1719.520	964.448	Ideal behavior	—
$\lambda_{21} - \lambda_{22}$ (J mol ⁻¹)	1791.088	45.014	1324.776		—
Deviation					
$ \Delta T _{av}$ (K) ^a	0.23	1.75	0.69	0.44	1.06
$ \Delta y_1 _{av}$ ^b	0.006	0.001	0.002	0.003	0.011
$ \Delta y_2 _{av}$ ^b	—	—	—	—	0.015
NRTL Parameters					
$g_{12} - g_{22}$ (J mol ⁻¹)	4193.718	1985.986	5605.956	Ideal behavior	—
$g_{21} - g_{11}$ (J mol ⁻¹)	−2756.582	−388.600	−3105.460		—
α_{12}	0.3	0.3	0.3		—
Deviation					
$ \Delta T _{av}$ (K) ^a	0.23	1.70	0.55	0.44	0.70
$ \Delta y_1 _{av}$ ^b	0.006	0.001	0.001	0.003	0.011
$ \Delta y_2 _{av}$ ^b	—	—	—	—	0.019
UNIQUAC Parameters					
$u_{12} - u_{22}$ (J mol ⁻¹)	1189.679	−377.202	243.434	Ideal behavior	—
$u_{21} - u_{11}$ (J mol ⁻¹)	−793.533	1176.247	472.084		—
Deviation					
$ \Delta T _{av}$ (K) ^a	0.20	1.77	0.70	0.44	1.46
$ \Delta y_1 _{av}$ ^b	0.006	0.001	0.002	0.003	0.013
$ \Delta y_2 _{av}$ ^b	—	—	—	—	0.024
Modified UNIFAC (Dortmund)					
Deviation					
$ \Delta T _{av}$ (K) ^a	0.29	1.34	0.83	0.42	1.43
$ \Delta y_1 _{av}$ ^b	0.006	0.001	0.003	0.003	0.012
$ \Delta y_2 _{av}$ ^b	—	—	—	—	0.013
ASOG					
Deviation					
$ \Delta T _{av}$ (K) ^a	0.25	0.45	0.99	0.49	1.35
$ \Delta y_1 _{av}$ ^b	0.006	0.001	0.003	0.004	0.011
$ \Delta y_2 _{av}$ ^b	—	—	—	—	0.013

^a $|\Delta T|_{av} = \sum_{k=1}^{NDP} |T_{exptl} - T_{calcd}|_k / NDP$. ^b $|\Delta y_i|_{av} = \sum_{k=1}^{NDP} |y_{i,exptl} - y_{i,calcd}|_k / NDP$, where NDP is the number of data points.

of these models, and good agreement was found between the experimental and predicted VLE. Two group contribution methods, i.e., the modified UNIFAC (Dortmund) and ASOG, were also valid methods for predicting the VLE behaviors for the ternary and binary systems investigated in this work, although the ASOG showed slightly poorer prediction accuracy for some of the systems. The experimental VLE values obtained in this work, as well as the predictions using the group contribution methods, will be useful for the design of hybrid configurations using liquid–liquid extraction and distillation alternatives for biobutanol purification using 2-ethyl-1-hexanol as the extractant for the liquid–liquid extraction process.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jced.4c00374>.

Description of the evaluation procedure for the standard uncertainties of pressure P , temperature T , and mole fraction of the liquid and vapor phase, x_i and y_i (PDF)

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Notes

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